TRANSISTORS AND AMPLIFIERS

CHAPTER 5

T3YBGT

வாரி பெருக்கி வளம்படுத்து உற்றவை
ஆராய்வான் செய்க வினை. – குறள் 512

விளக்கம்:
பொருள் வரும் வழிகளைப் பெருக்கச் செய்து, அவற்றால் வளத்தை உண்டாக்கி, வரும் இடையூறுகளை ஆராய்ந்து நீக்க வல்லவனே செயல் செய்ய வேண்டும்.

English Couplet 512:

Who swells the revenues, spreads plenty o'er the land,
Seeks out what hinders progress, his the workman's hand.

Couplet Explanation:
Let him do (the king's) work who can enlarge the sources (of revenue), increase wealth and considerately prevent the accidents (which would destroy it).

CONTENT

5.1. Introduction
5.2. Transistor
5.3. Bipolar junction transistor
5.4. Some facts about the transistor.
5.5. Transistor - Testing.
5.6. Transistor – Biasing.
5.7. Working of NPN - PNP transistor.
5.8. Operating modes of transistor.
5.9. Transistor configurations (CB, CE, CC)
5.10. Multistage transistor amplifier.
5.11. Voltage amplifier and power amplifier.
5.12. Classification of power amplifiers.
5.13. Feedback in amplifiers.
5.14. Distortion in amplifiers.
5.15. Transistor applications.

LEARNING OBJECTIVES

After the completion of this Chapter, the student will

1. Understand the basic structure of the bipolar junction transistor (BJT)
2. Working principles of PNP and NPN transistor
3. Know the transistor configuration as an amplifier (or) a switch
4. Difference between voltage amplifier and power amplifier.
5. Construction and configurations of transistors (common base, common emitter, common collector)
6. Application of transistors

> Try not to become a man of success, but rather try to become a man of value.
>
> – Albert Einstein

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